

Experiment G — Quadrature Speed/Equivariance Pareto

What is the right way to compare DH vs GL? **Same equivariance error**, then ask how much wall-clock time each method needs to reach it.

Setup (fairchem default EquiformerV2, lmax=6, mmax=2)

Real EquiformerV2Backbone from fairchem: 12 layers, 128 channels, 24 S2Activation modules. NVIDIA A100-SXM4-40GB, idle.

These are the function-signature defaults of EquiformerV2 in fairchem (`num_layers=12`, `sphere_channels=128`, `lmax_list=[6]`, `mmax_list=[2]`); the published OC20 SOTA model is larger (20 layers, lmax=8, mmax=8, ~150M params), so the +31% savings here is a lower bound for the production model. The 24 modules = $2 \times \text{num_layers}$ (one S2 act in graph attention + one in FFN per TransBlockV2).

Two measurements at the same (lmax, mmax)

Equivariance error of one S2-Act kernel (random coefficients, 5 rotations $\times$ 10 inputs); SiLU activation. *Source: verify_gl_grid.py.*

Wall-clock for one full forward (QM9, batch=8). Protocol: 5 independent runs per config; each run uses 10 warmup + 30 measured forwards on *distinct* QM9 batches drawn without replacement (no batch reuse within a run, fresh RNG seed per run). `cuda.sync()` around every measurement; fresh model construction per (run, config). 95% CI is reported as $t_{n_{runs}-1, 0.975} \times \text{s.e.}(\text{run_means})$ with $n_{runs} = 5$ as the unit of replication. *Source: verify_speedup_v2.json.*

Combined results

Config	Pts	Equiv err	Fwd (ms, 95% CI)
DH default	70	4.43×10^{-1}	113.50 ± 1.87
DH 2×	784	3.27×10^{-1}	164.66 ± 1.62 (+45.1%, $p < 10^{-4}$)
GL min	91	4.29×10^{-1}	112.68 ± 1.87 (−0.7%, n.s.)
GL match-DH	182	3.28×10^{-1}	113.17 ± 2.25 (−0.3%, n.s.)
GL 2×	392	3.28×10^{-1}	112.61 ± 2.35 (−0.8%, n.s.)

Matched-equivariance comparison

For target equiv err = 3.28×10^{-1} (the saturation floor at this mmax cropping):

- Cheapest DH: **DH 2×** at **164.66 ms** (DH default at 70 pts only reaches 4.43×10^{-1} , not enough)
- Cheapest GL: **GL match-DH** at **113.17 ms**
- **Savings:** 51.5 ± 2.36 ms = **−31.3% time**, $p < 10^{-4}$ from Welch’s t-test on $n = 5$ run means

So: at matched equivariance accuracy, GL saves ~31% wall-clock per forward.

Equivalently: GL achieves DH 2×’s equivariance at the same wall-clock as DH default (the GL-vs-DH-default differences are statistically zero). The +51.5 ms premium that DH pays to reach the equivariance floor goes away under GL.

Caveats

1. Result is for this specific (lmax=6, mmax=2) config and this GPU. Different model depths or hardware will scale it.
2. Equivariance is measured on *one* S2-Act kernel with random inputs, not on a trained model's intermediate features. Trained-model features are smoother (absolute equiv numbers will be much lower); the *ranking* between quadratures is what is claimed to transfer. See `dropin_pretrained.json` for validation on actual 4-layer trained checkpoints.
3. The kernel-level equivariance gap between GL match-DH and DH $2\times$ is within rounding (3.28×10^{-1} vs 3.27×10^{-1}).
4. The mmax=2 cropping caps achievable equivariance at ~ 0.328 regardless of grid; lower error needs mmax=lmax (architectural change). At lmax=mmax both methods can reach a $\sim 4\times 10^{-3}$ floor; there GL needs roughly half the latitude points of DH (see `verify_gl_grid.py`).
5. The wall-clock benchmark above does NOT load pretrained weights; it times architecture-level forward cost. The cost depends only on architecture and tensor shapes, not on weight values.

Reproduce

```
module load pytorch/2.6.0-1
# Equivariance for all configs at lmax=6, mmax=2
python3 experiments/verify_gl_grid.py
# Wall-clock: rigorous IID protocol, 5 runs x 30 distinct batches
PYTORCH_CUDA_ALLOC_CONF=expandable_segments:True \
python3 experiments/expG_verify_speedup_v2.py \
--batch_size 8 --n_runs 5 --n_iter 30 --n_warmup 10
# Drop-in test on actual trained checkpoints (small model)
PYTORCH_CUDA_ALLOC_CONF=expandable_segments:True \
python3 experiments/expG_dropin_pretrained.py
```